Determinant of square matrix of order n, $n \geq 4$

The concept of determinant can be extended to the case of square matrix of order $n, n \geq 4$. Let $A = [a_{ij}]_{n \times n}, n \geq 4$.

If we delete the ith row and j^{th} column from the matrix of $A = [a_{ij}]_{n \times m}$, we obtain a determinant of order $(n-1)$, which is called the **minor of the element** a_{ij}. We denote this minor by M_{ij}. The cofactor of the element a_{ij} is defined as $A_{ij} = (-1)^{i+j} M_{ij}$.

Result 7.2

For a given square matrix $A = [a_{ij}]_{n \times n}$ of order n, the sum of the products of elements of the first row with their corresponding cofactors is the determinant of A. That is,

$|A| = a_{11}A_{11} + a_{12}A_{12} + ... + a_{1n}A_{1n} = \sum_{j=1}^{n} a_{1j}A_{1j}$ which, by the definition of cofactors and minors, is same as

$$|A| = \sum_{j=1}^{n} (-1)^{1+j} a_{1j} M_{1j},$$

where A_{1j} denotes the cofactor of a_{1j} and M_{1j} denotes the minor of a_{1j}, j = 1, 2, ..., n. ■

Note 7.7

(i) If $A = [a_{ij}]_{n \times n}$ then determinant of A can also be denoted as det(A) or *det A* or Δ.

(ii) It can be computed by using any row or column.

7.3.2 Properties of Determinants

We can use one or more of the following properties of the determinants to simplify the evaluation of determinants.

Property 1

The determinant of a matrix remains unaltered if its rows are changed into columns and columns into rows. That is, $|A| = |A^T|$.

Since the row-wise expansion is same as the column-wise expansion, the result holds good.

Property 2

If any two rows / columns of a determinant are interchanged, then the determinant changes in sign but its absolute value remains unaltered.

Verification

$$\text{Let } |A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$= a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

$$\text{Let } |A_1| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \\ a_2 & b_2 & c_2 \end{vmatrix} \quad (\text{since } R_2 \leftrightarrow R_3)$$

$$= a_1(b_3c_2 - b_2c_3) - b_1(a_3c_2 - a_2c_3) + c_1(a_3b_2 - a_2b_3)$$

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